



CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: UniVerse [All-in-one Digital
Universe for University Students]

Group No: 08, CSE471 Lab Section: 18, Spring 2025	
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Submission Date:

Tech Stack:

Language: TypeScript

Framework: SvelteKit (most likely)

Styling: TailwindCSS or Bootstrap

Database: Supabase PostgreSQL

ORM (Optional): Supabase Client Library or Drizzle ORM

Deployment: Vercel or Supabase Hosting

Functional requirements:

Module 1: Authentication, Student Access & Dashboard

1. [Member-1] A student can register only using a valid **BRACU email address, complete their profile** (name, student ID, department, semester), and **log in/out securely**.
2. [Member-1] A student can view and update their profile information, such as name, email & change password feature allows users to update their login credentials.
3. [Member-2] A student can filter food items by category, such as snacks, lunch, or drinks, to quickly find their preferred items from the menu.
4. [Member-3] A student can create listings for second-hand items they want to sell by entering a title, description, price, and category, and can also edit or delete those listings as needed.
5. [Member-3] A student can click on an item listing to view detailed information (e.g., description, price, seller details) and proceed with the purchase if desired.
6. [Member-4] A student can check the current availability status of books, identifying whether they are available, reserved by someone else, or currently borrowed.

Module 2: Task and Project Organization

1. [Member-1] A student can view system-generated notifications and reminders—such as study room bookings, upcoming events, or transaction confirmations—on a centralized dashboard.

2. [Member-2] A student can pre-order food for pickup at a selected time, reducing the need to wait in lines during peak hours, and can cancel their pre-orders within the allowed time frame if they change their mind or plans.
3. [Member-2] A student can search using keywords in the menu so that he/she can order their desired food quickly.
4. [Member-3] A student can hire a volunteer for part-time work in the marketplace
5. [Member-4] A student can reserve library books online without needing to physically visit the library.
6. [Member-4] A student can view their active reservations and borrowing history, helping them track overdue or borrowed materials.

Module 3: Dashboard and Notifications

1. [Member-1] A student can request a live banner advertisement, and the admin can accept or reject. The live banner will be on the landing page like a billboard.
2. [Member-2] A student can rate food items on their quality, taste, or presentation.
3. [Member-2] A student can suggest new food items or dietary options, ensuring the cafeteria accommodates various tastes and dietary needs
4. [Member-3] A student can mark an item as sold after a successful transaction, making sure their listing does not appear in search results anymore.
5. [Member-3] A student can save items they're interested in to a wishlist for future reference or purchase
6. [Member-4] A student can view their study through a digital calendar or timetable and make room live bookings for specific time slots and durations.
7. [Member-4] A student can request a study group based on their study or book interest.

Extra Module :

[Member-2]

1. A student can view the cafeteria menu for the current day or the entire week, helping them plan meals.
2. A student can check their past food orders through an order history page, which is useful for reordering favorites or tracking dietary habits.

[Member-3]

1. A student can browse all listings with filters based on item categories (like books, calculators, gadgets), which helps narrow down their search.
2. A student can rate or leave reviews for sellers after completing a transaction, helping build credibility within the platform.
3. A student can view a complete history of items they've bought or sold, allowing for easy reference or transaction tracking.

[Member- 1 & 4]

1. A student can search for books using the book title, author's name, or department filter to quickly locate academic resources.
2. A student can rate or write reviews on books they've used, providing useful feedback for other readers.
3. A student can edit or cancel their own room bookings if plans change, ensuring flexibility in scheduling.
4. A student can suggest new books that should be added to the library, contributing to the expansion of available resources
5. Library study group members can chat with each other